School Curriculum and Standards Authority

JUDGING STANDARDS IN YEAR 8

MATHEMATICS

Assessment pointers validate teachers' professional judgement when reporting against a five-point scale. The pointers:

- are examples of evidence in relation to the achievement standard
- should be used with the annotated student work samples
- exemplify what *students may demonstrate* rather than a checklist of *everything they should do*.

YEAR 8 MATHEMATICS ACHIEVEMENT STANDARD

Number and Algebra

At Standard, students solve everyday problems involving rates, ratios and percentages. They describe index laws and apply them to whole numbers. Students describe rational and irrational numbers. They solve problems involving profit and loss. Students make connections between expanding and factorising algebraic expressions. They use efficient mental and written strategies to carry out the four operations with integers. Students simplify a variety of algebraic expressions. They solve linear equations and graph linear relationships on the Cartesian plane.

Measurement and Geometry

Students solve problems relating to the volume of prisms. They make sense of time duration in real applications. Students identify conditions for the congruence of triangles and deduce the properties of quadrilaterals. They convert between units of measurement for area and volume. Students perform calculations to determine perimeter and area of parallelograms, rhombuses and kites. They name the features of circles and calculate the areas and circumferences of circles.

Statistics and Probability

Students model authentic situations with two-way tables and Venn diagrams. They choose appropriate language to describe events and experiments. Students explain issues related to the collection of data and the effect of outliers on means and medians in that data. They determine the probabilities of complementary events and calculate the sum of probabilities.

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YEAR 8 MATHEMATICS ASSESSMENT POINTERS

	A Excellent achievement	B High achievement	C Satisfactory achievement	D Limited achievement	E Very low achievement
Number and Algebra					
Number and Place Value	Efficiently uses the index laws to simplify numerical expressions with multiple steps involving positive, integral indices and the zero index and distributive index law.	Uses the index laws to simplify numerical expressions with multiple steps involving positive, integral indices, zero index law and distributive index law.	Explains the connection between the expansion of numbers expressed in positive, integral index form and the index laws. Apply the index laws to whole numbers. For example: $4^3 \times 4^2 = 4^{3+2} = 4^5$ $4^5 \div 4^2 = 4^{5-2} = 4^3$	Uses the expansion of numbers expressed in positive index form to determine simple products and quotients. For example: $4^3 \times 4^2 = 4 \times 4 \times 4 \times 4 \times 4$ $= 4^5$ $7^5 \div 7^3 = \frac{7\times7\times7\times7\times7}{7\times7\times7} = 7^2.$	Does not meet the requirements of a D grade.
Real numbers	Chooses and uses efficient mental and written strategies and appropriate digital technology to carry out the four operations with integers and rational numbers.	Chooses and uses efficient and appropriate mental and written strategies or digital technology to carry out the four operations with integers and simple rational numbers.	Uses efficient mental and written strategies and appropriate digital technology to carry out the four operations with integers.	Uses mental and written strategies and appropriate digital technology to carry out the four operations with integers.	Does not meet the requirements of a D grade.
	Uses numbers in exact form, including square roots and pi, in calculations and final solutions.	Uses reasoning to show approximate locations of pi and other familiar irrational numbers on a number line. Uses pi in exact form for calculations where an example has been provided.	Identify and justify classification of rational and irrational numbers.	Uses technology to identify simple fractions or decimals as being terminating, recurring or neither.	
	Efficiently solves authentic problems involving percentages, rates and ratios, which require multiple calculations, with and without digital technologies.	Solves authentic problems involving percentages, rates and ratios, which require multiple calculations, with and without digital technologies.	Solves simple authentic problems involving percentages, rates and ratios with and without digital technologies.	Solves simple, everyday one-step problems involving percentages, fractions, rates and ratios with and without digital technologies.	
Money and financial mathematics	Solves unfamiliar, multi-step authentic problems involving profit/ loss including inverse calculations with and without digital technologies.	Solves familiar, multi-step authentic problems involving profit/loss with and without digital technologies.	Solves simple authentic problems involving profit/ loss with and without digital technologies.	Solves familiar financial problems involving profit and loss with digital technologies.	Does not meet the requirements of a D grade.

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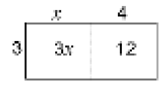
Patterns and algebra	Flexibly moves between factorised, expanded and simplified complex algebraic expressions, choosing the most appropriate form where variables are to the power of one.	Moves between factorised, expanded and simplified algebraic expressions where variables are to the power of one.	Moves between factorised and expanded algebraic expressions involving a single numerical common factor.	Uses a model to expand a binomial bracket using a single numeric factor. For example:  $3(x+4) = 3x + 12$	Does not meet the requirements of a D grade.
Linear and non-linear	Determines the rule for an authentic linear relationship. Graphs linear relationships using efficient methods including two points or slope and vertical intercept.	Completes a table of values to determine the rule for an authentic situation with a linear relationship, and plots a continuous graph from the resulting points on the Cartesian plane.	Completes a table of values from a simple linear rule or authentic situation and plots a continuous graph from the resulting points on a Cartesian plane.	Plots points from a given table of values on the Cartesian plane using an appropriate set of axes and scale.	Does not meet the requirements of a D grade.
	Uses algebraic techniques to solve linear equations related to authentic situations.	Uses graphs and algebraic techniques to solve related problems.	Uses tables of values and graphs of linear relationships to solve simple related problems.	Uses arithmetic methods to solve problems involving linear relationships.	
	Graphically and algebraically compares authentic linear relationships, such as total cost of a service charged over time t, given by functions. For example: Cost $(C) = 45t + 70$ or Cost $(C) = 50t + 40$ using precise language.	Interprets linear relationships related to simple authentic problems using some appropriate mathematical language.	Matches simple linear relationships to their graphs, comparing slope and vertical-axis intercept.	Identifies a given graph as linear or non-linear.	
	Solves multi-step linear equations including those involving brackets on both sides of the equation. Gives detailed reasoning and verifies the solution.	Algebraically solves linear equations involving brackets on one side of the equation.	Solves linear equations sometimes involving brackets or simple rational numbers. Shows working and verifies the solution.	Solves two-step linear equations, without justification of working.	

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Measurement and Geometry					
Using units of measurement	Ensures consistency of area and volume units when solving multistep problems.	Recognises mixed units in measurement problems and converts them to a common unit when calculating a perimeter, area or volume.	Converts given lengths, areas and/or volume from one unit to another.	Converts given lengths from one unit to another.	Does not meet the requirements of a D grade.
	Consistently expresses appropriate linear, area and volume units in solutions.	Expresses answers to measurement problems with appropriate units.	Expresses answers with linear units and sometimes area or volume units as appropriate.	Expresses answers to problems involving measurement, but usually omits the units.	
	Extracts information from a given diagram showing dimensions of a composite shape which include one or more of the following; parallelogram, rhombus, trapezium or kite. Solves complex problems involving the perimeter and/or area including the inverse use of simplified formulas after the substitution of values.	Extracts information from a given diagram showing the dimensions of a parallelogram, rhombus, trapezium or kite, and solves simple problems involving the perimeter and/or area.	Calculates the perimeter and/or area of a parallelogram, rhombus or kite, given a diagram showing relevant dimensions.	Calculates the perimeter of a quadrilateral, given a diagram showing all its dimensions.	
	Determines the radius and diameter of circles, given their circumference or area and the perimeter and area of composite figures involving circles, quadrants or semicircles.	Determines the circumference, perimeter and area of circles and simple composite figures involving circles, quadrants and semicircles.	Chooses appropriate formulas to calculate the circumference, in terms of diameter or radius, and area of circles in routine or familiar problems.	Uses formulas to calculate the circumference and area of circles, but sometimes confuses radius and diameter or circumference and area.	
	Solves a variety of problems involving circles and part circles, giving approximate or exact answers as required.	Plans and carries out the solution to practical problems involving circles, quadrants or semicircles, including those which involve more than one step.	Solves simple and familiar practical problems involving circles.	Attempts solutions to some practical problems involving circles, but confuses situations where circumference or area are involved.	

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Using units of measurement	Identifies and accounts for specific or changed conditions when solving problems in measurement contexts.	Identifies and accounts for some specific or changed conditions when solving problems in measurement contexts.	Identifies relevant information from a simple diagram or straight forward text when solving practical measurement problems involving limited steps.	Identifies some relevant information from a simple diagram or straight forward text when solving practical measurement problems involving one step.	Does not meet the requirements of a D grade.
	Solves a variety of problems involving the volume and capacity of shapes involving right prisms by considering the cross-section and the height of layers of this cross-section.	Selects and uses the appropriate formula to determine the volume of rectangular and triangular prisms and prisms in general where the area of the base and height is given.	Links the cross-section of a right prism to the base of the prism and uses the area of the base and the height of the prism to calculate the volume of rectangular or triangular prisms.	Calculates the volume of a rectangular prism given a diagram labelled with dimensions.	
	Solves complex problems involving time duration in real life applications, such as planning a 24-hour TV schedule with multiple constraints.	Solves multi-step problems involving time duration in real-life applications, such as planning a trip to town to watch a movie and arriving back home in time for tea.	Calculates time duration in simple real-life applications, using 12- and 24-hour time within a single time.	Calculates time duration for familiar applications such as calculating the time between buses from a local bus timetable.	
Geometric reasoning	Uses transformations to verify and communicate the conditions of congruency of triangles in a composite diagram.	Identifies the presence, or not, of a pair of congruent triangles in a set of triangles of varying orientations, and names the appropriate conditions for congruency.	Identifies the congruent pairs of triangles in a similar orientation and communicates the appropriate conditions of congruency with some reference to sides and angles.	Identifies congruency in a pair of triangles in a diagram with the same orientation.	Does not meet the requirements of a D grade.
	Applies the correct conventions to set out the reasoning when calculating the value of a side or angle in a pair of congruent triangles contained in a composite shape.	Applies the properties of congruent figures to calculate the value of a missing side or angle in a pair of congruent triangles contained in a composite shape.	Completes information in a given diagram and calculates the value of a missing side or angle in a pair of congruent triangles.	Calculates the value of a missing side or angle in a pair of congruent triangles.	
	Applies the correct conventions to set out the reasoning when establishing that a given diagonal divides the rectangle, parallelogram or kite into a pair of congruent triangles.	Uses the geometric properties of the rectangle, parallelogram or kite to establish that a given diagonal divides the quadrilateral into a pair of congruent triangles.	Draws a diagonal of the rectangle, parallelogram or kite to divide the quadrilateral into a pair of congruent triangles and marks the congruent sides and angles.	Identifies and correctly names quadrilaterals.	

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Geometric reasoning	Solves problems of composite figures containing a rectangle, parallelogram, rhombus or kite using the properties of these figures and the congruent triangles included in them.	Proves and uses congruency of triangles in given quadrilaterals such as rectangle, parallelogram, rhombus or kite and calculates the value of sides or angles as required.	Calculates length or angles of a quadrilateral such as a rectangle, parallelogram, rhombus or kite given measurements and using congruency of triangles contained in the figures.		Does not meet the requirements of a D grade.
Statistics and Probability					
Chance	Applies probability to solve problems. Interprets the sample space using relevant language, including the distinction between inclusive 'or' (*A* or *B* or both) and exclusive 'or' (*A* or *B* but not both).	Constructs a two-way 2x2 table giving the number of a Venn diagram and calculates probabilities of compound events such as: {The person selected Is a boy in the debating club}.	Uses the information from a given Venn diagram or a two-way table to assign probabilities of compound events such as *'A and B'; 'A or B' 'A only'*. Uses the probability of event A to calculate its compliment *'Not A'.*	Assigns a probability value to a single event by counting possible outcomes in the relevant sample space or subset of a given Venn diagram.	Does not meet the requirements of a D grade.
Data representation and interpretation	Chooses between a Venn diagram and a two-way table to model the sample space of a given single-step experiment involving two events. Describes each region as a subset of the sample space and calculates the associated probability in multiple number forms.	Interprets given information to allocate the numbers of elements in the subsets of a Venn diagram including intersections where some sample points or elements belong to more than one subset. Draws and labels the Venn diagram showing information.	Constructs a two-way table from an experiment such as 'Rolling a die' AND 'Tossing a coin' to list all the possible outcomes such as (6,H).	Lists the outcomes of a single event as a sample space such as 'Rolling a die' implies {1,2,3,4,5,6}.	Does not meet the requirements of a D grade.
	Plans and carries through an appropriate method for the collection and display of categorical data for tasks such as '*Compare the methods of transport year 8 students use to get to school*'.	Identifies issues related to the collection of data such as 'Sample size' or 'Stratified sampling'.	Displays categorical data using a column graph with axes labelled appropriately.	Organises collection of raw data using tallies and frequency tables.	

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Data representation and interpretation	Calculates the mean and the median of a data set, explaining how the relationship between the two is influenced by outliers. Identifies issues related to the collection of data in a variety of situations and the variation of the mean and/or median between samples.	Uses a formal method such as a graph to identify an outlier. Collects repeated samples of numerical data from a given population, demonstrating and explaining possible variation of the mean and/or median between samples, including reference to sample size and outliers.	Identifies an outlier in a set of numerical data where a value is way off the scale. Compares the mean and median of repeated samples from a familiar situation, recognising the possible effects of outliers when making predictions about the population.	Calculates the mean and median of numerical data in a list. Gives a simple explanation for inaccuracies in collecting data in familiar situations such as when measurement or counting is involved.	Does not meet the requirements of a D grade.